

VU KHANH CAO

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EDUCATION

Carleton University, Ottawa, ON

Sept 2023 – Present

Bachelor of Computer Science (Honours) — Artificial Intelligence and Machine Learning

GPA: 3.75 / 4.00

SKILLS

Programming	C/C++, Python, Java, TypeScript
Web & Systems	React, Flask, Angular, Node.js, Express, Spring Boot
Data & AI	TensorFlow, PyTorch, Keras, OpenCV
DevOps & Tools	Linux, Azure, Docker, Kubernetes, ArgoCD
Databases	PostgreSQL, MySQL, MongoDB

EXPERIENCE

Software Developer

May 2026 – Sept 2026

Bank of Canada

Ottawa, ON

- Worked on modernizing secure, mission-critical systems supporting multi-billion-dollar transactions each year using Angular and Spring Boot.
- Designed and developed agentic tools to automate code review and verification within Azure CI/CD pipelines, ensuring code quality and security matching strict internal guidelines.

Software Developer

Jun 2025 – Dec 2025

Ovenns Analytics Inc.

Ottawa, ON

- Contributed to the design, development, and maintenance of a production-grade web platform managing food preferences for restaurant staff and customers.
- Built secure, scalable backend services using Node.js, Express, and PostgreSQL, supporting multi-tenant access and reliable data handling.

PROJECTS

GURT (Ground Unit Response Tank) - Best Hardware Hack, UOttHack 8

Jan 2026

- Designed and implemented a real-time robotic control system combining software, embedded hardware, and secure WebSocket-based communication for real-time video streaming and command execution.
- Developed a custom QNX 8 camera driver on Raspberry Pi 4, focusing on reliability, low-latency communication, and increasing video throughput by 10x.

JARVIS - 3D Scene Reconstruction, UOttHack 7

Jan 2025

- Designed and implemented a low-cost physical-to-digital pipeline that turns camera footage into web-ready 3D views using QNX, machine learning, and Gaussian Splatting.
- Deployed and maintained services using Docker and Linux-based systems, emphasizing reliability and reproducibility.

TRACY (Tennis Realtime Analysis Coaching) - 3rd Place, QHacks 2024

Feb 2024

- Developed a computer vision and machine learning system that analyzed monocular footage using OpenCV, TensorFlow, Keras, YOLOv3, and TrackNet to track player movement and ball trajectories.
- Processed video footage with FFmpeg and generated trajectory graphs, heat maps, gameplay statistics, and AI-powered performance summaries through a React and Flask application integrated with the OpenAI API.